

Multi-Objective Optimization: A Student's Quick Guide

What is Multi-Objective Optimization (MOO)?

Real problems involve multiple, conflicting objectives (e.g., accuracy vs. model size, precision vs. recall, energy vs. performance). The target is not a single “best” model but the Pareto front of trade-offs.

Core Concepts

Pareto dominance: A dominates B if it is no worse in all objectives and strictly better in at least one.

Pareto front: Non-dominated solutions representing optimal trade-offs.

Scalarization: Convert multiple objectives into one for standard optimization.

Basic Methods (Integrable with Deep Learning)

Weighted Sum Method

Equation: $L = \sum w_i f_i(\theta)$

Adjust weights to explore different trade-offs.

Paper: <https://link.springer.com/book/10.1007/978-1-4615-5563-6>

Code: https://pytorch.org/tutorials/beginner/blitz/neural_networks_tutorial.html#loss-function

ϵ -Constraint Method

Equation: Maximize primary objective, constrain others (e.g., FLOPs $\leq$ threshold).

Optimize one main objective while treating others as constraints.

Paper: <https://www.sciencedirect.com/book/9780486402584/multiobjective-decision-making>

Code: <https://github.com/paul-grabowski/fairtorch>

Chebyshev (Tchebycheff) Scalarization

Equation: $L = \max_i w_i |f_i(\theta) - z_i^*|$

Balances progress across objectives, reduces bias from scaling.

Paper: <https://doi.org/10.1007/BF02788966>

Code: <https://github.com/isl-org/MOO-Bench>

Multi-Gradient Descent (MGDA / PCGrad)

Equation: Find descent directions that satisfy multiple objectives jointly.

Gradient-based method integrated into SGD/Adam.

Paper: <https://arxiv.org/abs/1810.04650>

Code: <https://github.com/isl-org/MGDA>, <https://github.com/tianheyu927/PCGrad>

Evolutionary/Population-Based Methods (NSGA-II)

Equation: Maintain population to explore Pareto front.

Explores diverse solutions, common in Neural Architecture Search.

Paper: <https://ieeexplore.ieee.org/document/996017>

Code: <https://github.com/anyoptimization/pymoo>, <https://github.com/DEAP/deap>